Percentage and Percentile (Detailed)

Percentage and **Percentile** are two important statistical concepts. Although they sound similar, they have **different meanings and applications**.

- **Percentage** tells **how much out of 100**.
- **Percentile** tells **how a value ranks compared to other values**.

Understanding the difference is essential in **Statistics, Data Science, Machine Learning, Competitive Exams, and Business Analytics**.

What is Percentage?

A **Percentage** is a way of expressing a number as a **fraction of 100**.

The word **percent** means **"per hundred."**

Definition

> **Percentage is a number or ratio expressed as a fraction of 100.**

Percentage Formula

$$
\text{Percentage}=\frac{\text{Part}}{\text{Whole}}\times100
$$

Example 1

A student scores **450 marks out of 500**.

Percentage

$$
\frac{450}{500}\times100=90\%
$$

Answer: 90%

Example 2

A company sells **800 products** out of **1000 produced**.

Percentage

[
\frac{800}{1000}\times100=80%
]

Answer: 80%

Example 3

There are **30 girls** in a class of **50 students**.

Percentage

[
\frac{30}{50}\times100=60%
]

Answer: 60%

Percentage Visualization

genui{"fractions_ratios_percentages_learning_block_staging":
{"type_id":"PERCENT_PART_WHOLE_PROPORTION"}}

Applications of Percentage

- Student examination results
- Discount calculations
- Profit and loss
- Tax calculation
- Election voting percentage
- Attendance percentage
- Business growth analysis

Advantages of Percentage

- Easy to understand.
- Allows comparison between different quantities.
- Widely used in daily life.
- Useful in reports and dashboards.

Limitations of Percentage

- Does not indicate a person's rank.
- Cannot describe the distribution of a dataset.
- Two people with the same percentage may have different rankings.

What is Percentile?

A **Percentile** indicates the **relative position (rank)** of a value within a dataset.

Definition

> **The p-th percentile is the value below which p% of the observations fall.**

For example:

- **25th percentile (P25)** → 25% of observations are below this value.
- **50th percentile (P50)** → Median.
- **75th percentile (P75)** → 75% of observations are below this value.

Simple Meaning

If your exam score is in the **90th percentile**, it means:

> **You scored higher than 90% of the students.**

It **does not** mean you scored **90% marks**.

Example

Suppose 100 students take an exam.

Your marks = **72**

Your percentile = **85**

This means:

- 85 students scored **less than or equal to 72**.
- 15 students scored **higher than 72**.

Your percentage could be **72%**, but your **percentile is 85**, because percentile depends on **how others performed**.

Percentile Formula (Rank-Based)

One common method is:

$$
\text{Percentile Rank}=\frac{\text{Number of values below the score}}{\text{Total number of values}}\times100
$$

Example

Marks of 10 students (sorted):

```
20, 35, 40, 50, 55, 60, 65, 75, 85, 95
```

Student's marks = **65**

Values below 65 = **6**

Total students = **10**

Percentile Rank

$$
\frac{6}{10}\times100=60
$$

Answer: The student's score is at approximately the **60th percentile**.

> **Note:** Different software packages (Excel, NumPy, Pandas, R, etc.) use slightly different formulas for calculating percentiles, so the exact value may vary slightly.

Common Percentiles

Percentile	Meaning
0th	Minimum value
25th (Q1)	First Quartile
50th (Q2)	Median

75th (Q3)	Third Quartile
100th	Maximum value

Percentiles in Data Science

Percentiles are widely used in:

1. Outlier Detection

Using the **Interquartile Range (IQR)**:

- Q1 = 25th percentile
- Q3 = 75th percentile

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2. Data Distribution

Understand how values are spread.

Example:

- P10
- P25
- P50
- P75
- P90
- P95

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3. Salary Analysis

Example:

> 90th percentile salary = ₹25,00,000

This means:

90% of employees earn less than ₹25 lakh.

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4. Healthcare

Doctors compare:

- Height percentile

- Weight percentile
- BMI percentile

Example:

A child in the **75th height percentile** is taller than **75%** of children in the reference group.

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5. Competitive Exams

Many entrance exams report **percentile** instead of percentage.

Examples:

- CAT
- JEE Main
- GRE
- GMAT

Example: Percentage vs Percentile

Suppose there are **100 students**.

Rahul scored **80 marks out of 100**.

Percentage

[

\frac{80}{100}\times100=80%

]

Percentage = **80%**

Now assume Rahul performed better than **95 students**.

His percentile is:

95th Percentile

Interpretation

- Percentage = **80% marks**
- Percentile = **Better than 95% of students**

Difference Between Percentage and Percentile

Feature	Percentage	Percentile
Meaning	Part out of 100	Relative rank in a dataset
Formula	(Part ÷ Whole) × 100	Based on position/rank in ordered data
Depends on Others?	❌ No	✅ Yes
Indicates	Performance based on total marks	Position compared to others
Range	0–100%	0th–100th percentile
Used In	Exams, finance, business	Competitive exams, statistics, data analysis

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Real-Life Examples

Example 1: School Exam

Marks = **450/500**

Percentage = **90%**

Percentile = **92**

Meaning:

- You scored **90% marks**.
- You performed better than **92% of students**.

Example 2: Height

Your height = **175 cm**

Height percentile = **80**

Meaning:

You are taller than **80%** of people in the comparison group.

Example 3: Salary

Salary = **₹10 lakh**

Salary percentile = **60**

Meaning:

You earn more than **60%** of employees.

Python Example

```
import numpy as np

marks = [55, 60, 65, 70, 75, 80, 85, 90]

print("50th Percentile:", np.percentile(marks, 50))
print("75th Percentile:", np.percentile(marks, 75))
print("90th Percentile:", np.percentile(marks, 90))
```

Output

```
50th Percentile : 72.5
75th Percentile : 81.25
90th Percentile : 86.5
```

Applications in Data Science

Percentage

- Accuracy of Machine Learning models
- Sales growth
- Conversion rates
- Profit margins
- Customer retention

Percentile

- Outlier detection
- Data preprocessing
- Feature engineering
- Salary benchmarking
- Performance evaluation
- Risk analysis

Summary

Percentage	Percentile
Expresses a value out of 100	Indicates the relative position in a dataset
Calculated using part and whole	Calculated using ordered data and rank
Does not depend on other	Depends on the distributio

observations	n of all observations
Example: 85% marks	Example: 85th percentile

Key Takeaway

- **Percentage** tells **"How much out of 100?"**
- **Percentile** tells **"Where do you stand compared to others?"**
- Two students can have the **same percentage** but **different percentiles** if they are compared against different groups.
- In **Data Science**, **percentiles** are especially useful for understanding data distributions, detecting outliers, and summarizing datasets, while **percentages** are commonly used to report proportions, rates, and model performance metrics.

Link : CHATGPT